

Industrial Greenhouse Monitoring and Forecasting System

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Abstract—In this paper, the way to create and implement a system that uses machine learning elements to monitor environmental conditions in a greenhouse is presented. The system includes a sensor block, a control block and a power block. The hardware component includes light, humidity and temperature sensors, Li-Ion battery and a special USB charging circuit. The Raspberry Pi system manages all these functions, and the data collected by the sensors will be taken by the system and stored in a database. The data will then be used to train an algorithm based on automatic learning for the automatic adjustments of the greenhouse. Therefore, the greenhouse can be controlled automatically, and the user will be notified by a remote e-mail alert system if certain limits are exceeded.

Keywords— *Raspberry Pi, sensors, machine learning (ML), artificial intelligence (AI).*

I. INTRODUCTION

Starting from the block diagram of the proposed system (Fig.1), we then designed the system and made the electrical diagram to have the support for the physical implementation of the project. For this step, the design software program Altium Designer was used. The Altium Designer software is used in the electronics industry to create, and design printed circuit boards (PCBs). This software provides an integrated environment that contains several tools for creating schematics, component placement, PCB routing, circuit simulation, and the creation of the documentation required for production. All aspects of the development process are covered by the integrated and unified environment provided by Altium Designer for the development of electronic products. The features it provides include front-end design and capture (allows users to place and connect components on the schematic, defining circuit relationships and functionality) and physical PCB design (allows printed circuit board design). Users can configure the physical layout of the PCB by placing components on the circuit board and drawing the connections between them. Using Altium Designer for printed circuit board (PCB) design and development has a few advantages, such as: an easy-to-understand interface, advanced capabilities, coordinated design environment, efficient administration of components, comprehensive simulation and analysis, integration of cloud services.

The automatic greenhouse checking and monitoring system consists of a complex set of components, with a

computer based on Raspberry Pi4. This system is designed to be powered directly from a fixed voltage source or from a Li-Ion battery, having the ability to charge this battery. From the hardware point of view, the system is designed to measure the battery level by means of an ADC built with discrete elements, different temperature, humidity, barometric pressure sensors and control relays for various functions.

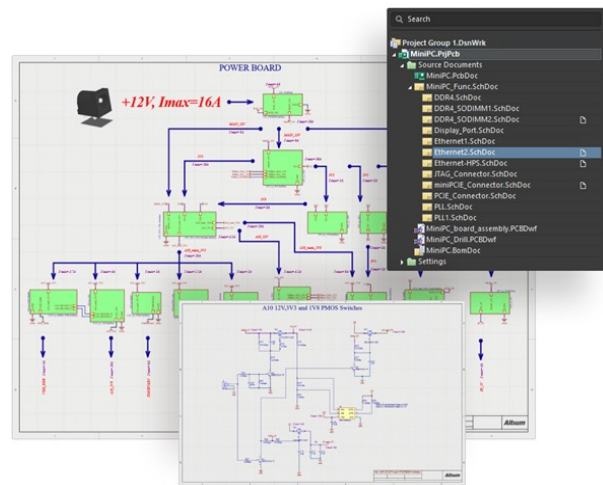


Fig. 1: Simulating a circuit in the interface of Altium

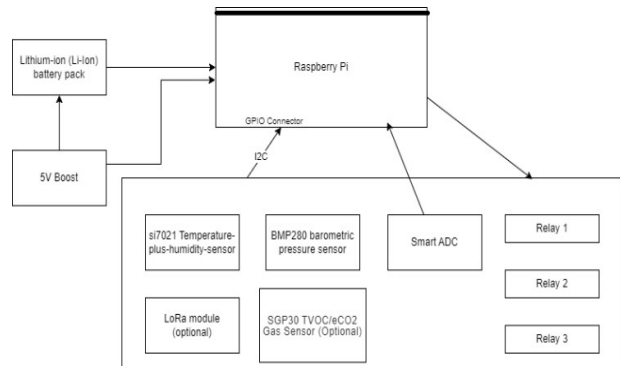


Fig. 2: Block diagram of the proposed system

II. METHODS AND DESIGN

In the greenhouse, regular measurements of temperature, light intensity, atmospheric pressure and humidity are required to track environmental conditions in order to make predictions, but also to trigger alarms when environmental parameters exceed the optimal limits set by the user.

Therefore, we made an air parameters monitoring system like this using a Si7021-A20 temperature and humidity sensor. It was soldered on the board with the soldering gun and then connected to the microcontroller according to the wiring diagram via the I2C interface, which is connected to physical pins 3 and 5, respectively the SDA (Serial Data) and SCL (Serial Clock) pins of the microcontroller. We also used the BMP280 sensor to collect atmospheric pressure data.

The core operating system consists of a Raspberry Pi 4 model B development board with a Linux operating system. Its role is to collect the data from the sensors mounted on the expansion board and to process them. The software implementation of the system was done using the Python3 language.

III. RESULTS

Following the practical realization and testing of the realized system, the objectives of this paper were achieved. The automatic learning algorithm was trained based on the data collected by the sensors with which the system is equipped, to automatically adjust the environmental parameters in the greenhouse. The system is able to automatically regulate environmental conditions, ensuring an optimal environment for plant growth and reducing the need for manual interventions.

As part of the work, we created an automatic system for monitoring and forecasting environmental parameters: humidity, temperature, light intensity and atmospheric pressure. After collecting these data, they were processed, we trained a neural network consisting of hidden layers, the first with 12 neurons and the second with 5 neurons.

The process presented in the block diagram, shown in the Fig. 3, starts with displaying the date, time and environmental parameters read from the sensors. Afterwards, a check of the system status is performed repeatedly at arbitrarily set time points (minute 30 and minute 59).

Depending on the environmental parameters, certain decisions and actions related to consumers are taken. With each iteration, sensor values are recorded to update the averaging calculation used in decision making.

Humidity check and associated actions: Checks if the humidity (H) is less than 25. If so, sends a warning email and displays a message. Activate the pump (PUMP) for 20 seconds to raise the humidity and then turn it off.

Temperature check and associated actions: checks if the temperature (T) is less than 10. If so, sends a warning email and displays a message. Activate the heating (HEAT) for 60 seconds to increase the temperature and then turn it off.

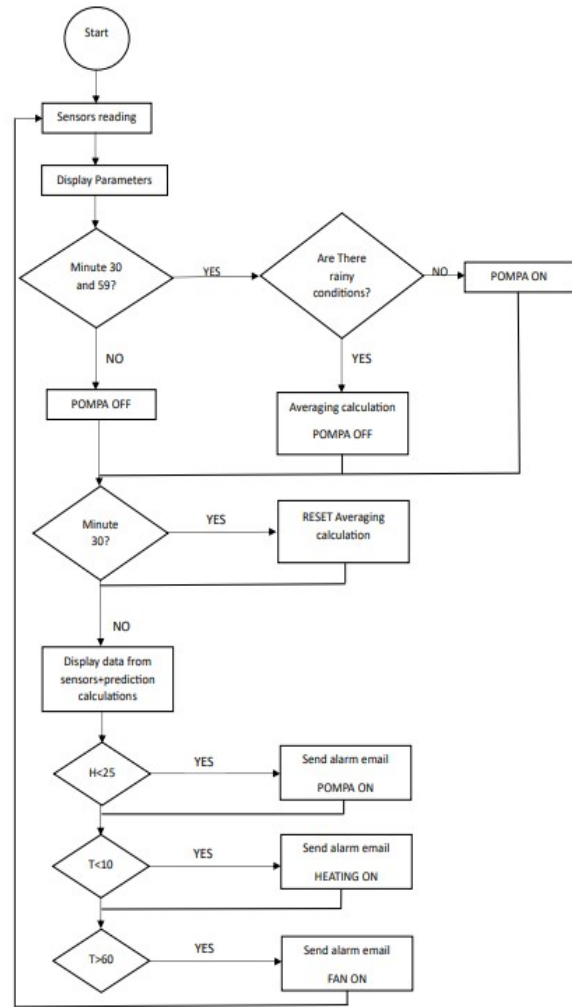


Fig. 3: Block diagram of the code

Checks if the temperature (T) is greater than 60. If so, sends a warning email and displays a message. Turn on the fan (FAN) for 60 seconds to lower the temperature and then turn it off.

It waits 4 seconds, after which the system resumes all this flow, until it is closed from the outside (system shutdown).

In the following images (fig. 4 and fig. 5) you can see the evolution of the environmental parameters obtained from the sensors, at the stage where we took real test data to train the neural network.

Finally, based on this system, we set according to our own preferences the limits that trigger alarms in real time and, at the same time, is able to make decisions in order to keep the environmental parameters in optimal conditions.

Following some test measurements, we obtained various results, depending on the environmental conditions in which they were carried out. To demonstrate the functionality of the system, we tested different scenarios, thus receiving the alert messages of the form "The humidity in the greenhouse is too low", a message we receive on the email address.

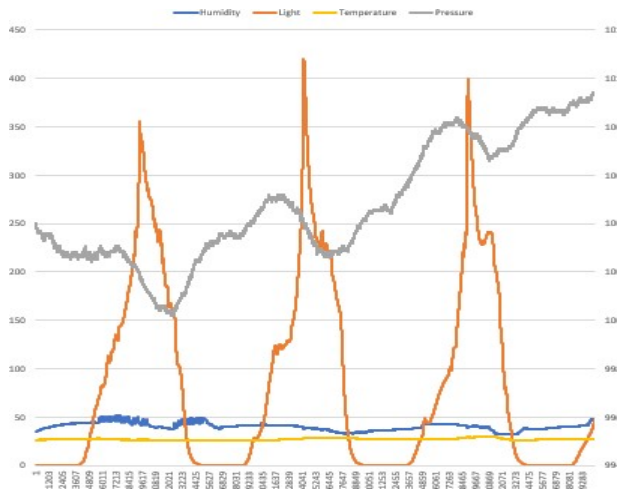


Fig. 4: Graphical evolution of environmental parameters

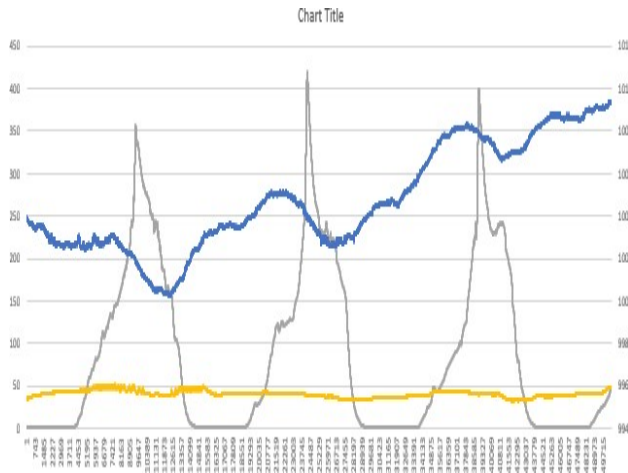


Fig. 5: Graphical evolution of environmental parameters

In the images below, one can see the results displayed in the terminal:

- Displaying the current date and time.
- Humidity and temperature display (SI7021 sensor).
- Temperature and pressure display (BMP280 sensor).
- ADC script execution: Displays a label for the raw light value and executes the content of the adc.py script, which performs specific operations for the analog-to-digital converter (ADC).
- Checking the current minute and actions associated with consumers.
- Environmental data recording and processing: it records the pressure values (P), humidity (H) and other data related to the light intensity (I) and Updates intensity integral by adding the new value for the averaging calculation based on which the system makes decisions.

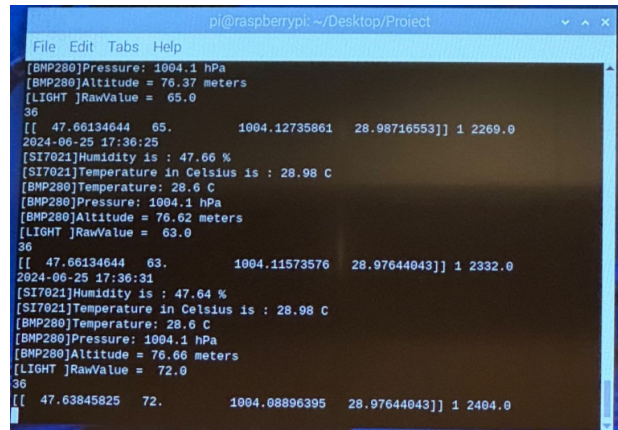


Fig. 6: Display environmental parameters read from the sensors

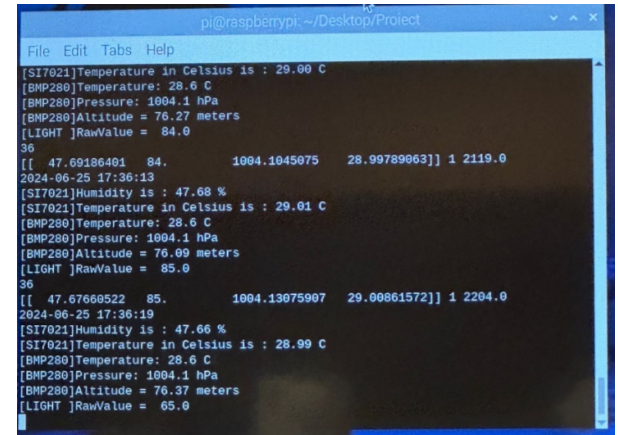


Fig. 7: Display environmental parameters read from the sensors

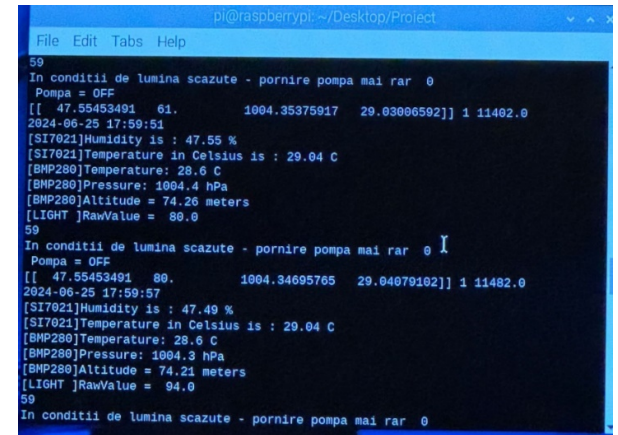


Fig. 8: Display environmental parameters read from the sensors

IV. CONCLUSIONS

The main goal of this article was to create a complete monitoring and forecasting system for industrial greenhouses, based on AI and Machine Learning technologies. We pursued the automation and optimization of environmental conditions in greenhouses to improve their management and maximize agricultural production. The developed system integrated sensors for monitoring environmental parameters, such as temperature, humidity and light, both inside and outside the greenhouse. These data were collected and stored in a centralized database, allowing

quick and efficient access for further analysis. The machine learning algorithm was trained on the basis of the collected data to forecast the energy consumption and to automatically adjust the environmental parameters in the greenhouse.

The system was able to automatically regulate environmental conditions, ensuring an optimal environment for plant growth and reducing the need for manual intervention. We have also implemented a protection and signaling system that sends alerts by e-mail if the environmental parameters exceed certain preset thresholds. This allows a quick intervention to prevent crop damage and to maintain optimal conditions in the greenhouse. In terms of precision and efficiency, the system has demonstrated the ability to automatically monitor and adjust environmental conditions, providing consistent and reliable results. The use of AI and ML allowed the system to adapt and learn from the collected data, constantly improving the accuracy of forecasts and the efficiency of automatic adjustments. A major advantage of the developed system is its scalability, allowing the connection and integration of several sensor nodes and devices in a centralized network. This makes it possible to adapt the proposed system to greenhouses of different sizes and complexities, offering a flexible and efficient solution for the management of industrial greenhouses.

In conclusion, the implementation of this monitoring and forecasting system for industrial greenhouses represents an important step towards automation and optimization of modern agriculture. The use of AI and ML technologies not only improves operational efficiency and the quality of agricultural production, but also contributes to sustainability

and responsible resource management. This work demonstrates the immense potential of advanced technologies in agriculture and paves the way for the development of innovative solutions to meet the challenges and future needs of this vital sector.

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